

Department of Computer Science & Engineering
Global Campus, Jakkasandra Post, Kanakapura Taluk, Ramanagara District, Pin Code: 562 112

PROJECT CENTRIC LEARNING (PCL) PHASE 2

				PROJECT TEAM ID	CoEPCL-0070
PROJECT TITLE		DIGITAL PAYMENTS THROUGH VOICE COMMAND			
TEAM MEMBERS					
S. No.	USN No.	NAME OF THE STUDENT	CLASS	MOBILE NO.	
1	23BTRCL192	I Dhanush	CSE-AIML	8328592840	
2	23BTRCL202	K Jahnavi	CSE-AIML	8688397124	
3	23BTRCL201	K Balaram	CSE-AIML	9392761604	
4	23BTRCL198	K Ravi Sankar	CSE-AIML	7989728125	
5	23BTRCL197	K Kundan	CSE-AIML	9059919256	
6	23BTRCL243	G Jathin	CSE-AIML	9963767225	
DETAILS OF THE PROJECT GUIDE					
NAME		S. Srinivas Kumar			
DESIGNATION		Asst. Professor			
DEPARTMENT		CSE-AIML			
E-Mail ID		srinivaskumar@jainuniversity.ac.in			
Phone No.		7993462416			
DOMAIN OF THE PROJECT					
Artificial Intelligence & FinTech					

PROJECT GUIDE

PROJECT COORDINATOR

ABSTRACT

Digital payment systems have become an essential part of modern financial transactions due to their speed, convenience, and accessibility. However, traditional payment platforms still require users to manually navigate applications, enter recipient details, input amounts, and authenticate transactions through PINs or passwords. This process can become inconvenient, especially for elderly users, visually impaired individuals, and people unfamiliar with digital interfaces.

To address these challenges, this project introduces a **Voice-Based Digital Payment System** that enables users to perform secure transactions through voice commands. The system converts spoken input into text, processes the payment intent using Natural Language Processing, verifies the speaker's identity through voice biometrics, and executes the transaction securely using Razorpay integration.

To further improve security, the system includes fraud detection and risk-adaptive authentication mechanisms. Suspicious or high-risk transactions require additional PIN verification before payment completion. This project combines accessibility, convenience, and security into one intelligent payment platform, demonstrating the practical use of Artificial Intelligence and Machine Learning in FinTech applications.

SCOPE OF THE PROJECT

- To build an intelligent voice-based payment system for digital transactions.
- To understand and process user payment commands through speech recognition.
- To verify the user's identity using voice biometric authentication.
- To implement fraud detection and risk analysis for transaction safety.
- To integrate Razorpay API for secure online payment processing.
- To improve accessibility for elderly and visually impaired users.
- To reduce the number of manual steps involved in digital payments.

EXISTING SYSTEM

Nowadays, most digital payments are completed through mobile banking applications, UPI apps, and digital wallet platforms. These systems require users to manually interact with their devices by opening applications, selecting recipients, entering transaction amounts, and confirming payments through PINs or OTP verification. Although these methods are widely used and reliable, they often involve multiple steps for a single transaction.

For elderly individuals, visually impaired users, and people unfamiliar with smartphones, navigating multiple screens and payment options can be confusing and difficult. While voice assistants exist in the market, they rarely support direct money transfers due to security concerns and lack of robust authentication mechanisms.

Problems in Existing System:

- Too many manual steps required for one transaction.
- Difficult for elderly and visually challenged users.
- Requires physical interaction with mobile devices.
- Limited voice assistant support for financial payments.
- Basic authentication methods only (PIN/OTP).
- Vulnerable to fraud and password theft.

Suggested Image:

- Screenshot/mockup of normal payment apps workflow
- OR simple diagram showing:
Open App → Select Contact → Enter Amount → PIN → Pay

PROBLEM IDENTIFICATION

Current digital payment systems lack simplicity and accessibility for many users. To complete even a simple transaction, users must go through multiple steps including opening the application, selecting contacts, entering payment details, and confirming the transaction manually. While regular smartphone users may find this manageable, it can become difficult for users with limited technical knowledge or physical disabilities.

Voice-based assistants partially solve the convenience problem, but most do not support secure financial transactions due to concerns regarding spoofing, voice cloning, and unauthorized access. A recorded or fake voice can potentially misuse the system if proper verification is not implemented.

Key Problems Identified:

- Lack of hands-free payment systems.
- No intelligent voice-based financial assistant available.
- Risk of spoofed or cloned voice attacks.
- Poor accessibility for elderly and disabled users.
- No adaptive authentication based on transaction risk.

PROPOSED SYSTEM

To solve the limitations of traditional payment systems, our project proposes a **Voice-Based Digital Payment System** that allows users to initiate financial transactions simply by speaking commands. The user can provide instructions such as recipient name and amount through voice, and the system processes the request automatically.

The system first converts voice input into text using speech recognition technology. After transcription, Natural Language Processing is used to extract important payment details such as recipient name, payment amount, and transaction intent. The speaker's voice is then verified against enrolled voice samples to confirm identity.

To ensure transaction safety, the system includes a risk-adaptive fraud detection mechanism. If a transaction is considered suspicious, the system requests additional PIN verification before proceeding. Once verification is complete, the transaction is processed through Razorpay integration.

Features of Proposed System:

- Voice-based transaction initiation.
- Automatic speech-to-text conversion.
- NLP-based payment command understanding.
- Speaker verification using voice biometrics.
- Fraud detection using machine learning.
- PIN verification for risky transactions.
- Razorpay payment gateway integration.

Suggested Image:

- **Main proposed workflow:**

Voice Input → STT → NLP → Verification → Fraud Check → Payment

LITERATURE SURVEY

Voice-based systems and artificial intelligence have seen significant growth in recent years, especially in the domains of smart assistants, automation, and financial technology. Several researchers have explored the use of speech recognition, Natural Language Processing (NLP), and machine learning for creating intelligent systems that can understand and process human voice commands effectively.

Studies in speech recognition have shown that modern Speech-to-Text models such as Whisper and deep learning-based Automatic Speech Recognition systems can accurately transcribe speech even in noisy environments and varying accents. These advancements have improved the reliability of voice-driven applications and enabled practical implementation in real-world systems.

Similarly, research in speaker verification has introduced deep learning architectures such as ECAPA-TDNN for biometric voice authentication. These models are capable of extracting speaker-specific features and verifying identity with high accuracy. In financial technology, fraud detection algorithms such as Isolation Forest and Random Forest have been used extensively to detect suspicious behaviour and prevent fraudulent transactions.

Key Findings from Literature:

- Speech recognition systems have improved significantly with deep learning.
- NLP helps machines understand human intent from natural language.
- Voice biometrics can provide an additional authentication layer.
- Fraud detection models can identify suspicious payment behaviour.
- Existing research supports combining AI and security for financial systems.

Suggested Image:

- Comparison table image / research comparison chart.

Example:

Author	Technology	Application
--------	------------	-------------

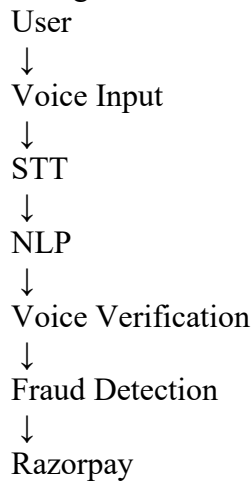
ARCHITECTURE & METHODOLOGY

The architecture of the proposed system is designed in a layered manner to ensure smooth processing, modularity, and secure transaction handling. The system begins with capturing voice input from the user and ends with payment execution after verification and risk analysis.

The architecture follows a three-layer model consisting of the Analysis Layer, Security Layer, and Transaction Layer. Each layer is responsible for a specific task in the payment workflow.

Suggested Image:

- Big architecture block diagram:



ALGORITHMS

The project uses multiple algorithms and machine learning models to perform different tasks within the system. These algorithms help improve accuracy, security, and overall intelligence of the payment platform.

Algorithm 1: Speech-to-Text Processing

This algorithm converts spoken user input into text using a pre-trained speech recognition model.

Steps:

1. Capture audio signal from microphone.
2. Preprocess the audio signal.
3. Pass input into speech recognition model.
4. Generate transcribed text output.

Algorithm 2: NLP Intent Detection

This algorithm determines the payment action from the user's transcribed command.

Steps:

1. Tokenize text input.
2. Analyze sentence structure.
3. Extract amount and recipient entities.
4. Predict payment intent class.

Algorithm 3: Voice Verification

This algorithm verifies the speaker's identity using stored voice biometric embeddings.

Steps:

1. Extract voice embeddings from live audio.
2. Compare with enrolled speaker embeddings.
3. Calculate similarity score.
4. Approve or reject speaker based on threshold.

Algorithm 4: Fraud Detection

This algorithm determines whether a transaction appears suspicious.

Steps:

1. Extract transaction-related features.
2. Pass features into fraud detection model.
3. Compute fraud probability score.
4. Classify as low, medium, or high risk.

RESULTS

The developed system was tested under multiple scenarios to evaluate its performance, reliability, and transaction security. Experimental testing demonstrated that the system successfully recognized voice commands, extracted transaction details accurately, and processed transactions with minimal delay.

The speaker verification module performed efficiently in distinguishing enrolled users from unauthorized speakers. Fraud detection mechanisms also successfully identified abnormal transaction patterns and triggered additional authentication when necessary.

Observed Results:

- High speech recognition accuracy under normal conditions.
- Accurate extraction of payment amount and recipient name.
- Successful voice authentication for enrolled users.
- Effective fraud detection for suspicious transactions.
- Razorpay integration successfully processed test payments.

Suggested Images:

- Accuracy graph/bar chart
- Confusion matrix
- Performance metrics chart

Use your conference paper graphs for inspiration.

TESTING

Testing was conducted to validate the performance of each individual module as well as the integrated end-to-end system. Different testing strategies were used to ensure reliability, security, and functionality.

Types of Testing Performed:

1. Unit Testing

- Tested each module independently.
- Verified correct input/output behaviour.

2. Integration Testing

- Tested communication between modules.
- Ensured proper data transfer between components.

3. System Testing

- Tested the entire application as one unit.
- Verified full payment workflow.

4. Unit Testing

- Tested each module independently.
- Verified correct input/output behaviour.

5. Integration Testing

- Tested communication between modules.
- Ensured proper data transfer between components.

CONCLUSION

The Voice-Based Digital Payment System successfully demonstrates that digital financial transactions can be securely and efficiently executed through voice commands. By integrating speech recognition, Natural Language Processing, voice biometrics, fraud detection, and payment gateway APIs, the project creates a smart payment platform that improves convenience and accessibility.

The system effectively reduces manual interaction during payment processing while maintaining transaction security through multi-layer authentication and fraud detection. This makes the platform suitable for elderly users, visually impaired individuals, and users seeking hands-free transaction experiences.

Overall, the project proves that combining Artificial Intelligence and Machine Learning with FinTech applications can significantly improve the future of digital payments and financial accessibility.

FUTURE SCOPE

The current system provides a strong foundation for voice-based financial transactions, but several improvements can be made in future versions to enhance performance and expand usability.

Future Enhancements:

- Add multilingual voice command support.
- Improve anti-spoofing with deepfake voice detection.
- Reduce latency through model optimization.
- Deploy the system on mobile applications.
- Add support for multiple payment gateways.
- Integrate blockchain-based transaction logging.
- Implement cloud-based distributed fraud detection.

REFERENCES

1. Desplanques, B., Thienpondt, J., and Demuynck, K., "ECAPA-TDNN: Speaker Verification Architecture."
2. Radford, A., et al., "Whisper: Robust Speech Recognition via Large-Scale Supervision."
3. Liu, F. T., et al., "Isolation Forest for Anomaly Detection."
4. Chen, T., and Guestrin, C., "XGBoost and Tree-Based Fraud Detection Models."
5. Razorpay API Documentation.
6. OpenAI Whisper Documentation.
7. SpeechBrain Speaker Verification Documentation.

